

# Alex Rivera

alex.rivera@example.com | (555) 123-4567 | linkedin.com/in/alexrivera-rtl

Target Role: RTL Design Engineer

## TECHNICAL SKILLS

Languages: Verilog, SystemVerilog, C, Python, Tcl, Bash

Hardware & EDA Tools: Vivado, Quartus, ModelSim, Synopsys Design Compiler, Cadence Genus

Protocols & Architectures: AXI4, APB, UART, SPI, I2C, Clock Domain Crossing (CDC), FSM, FIFOs, Static Timing Analysis

## EDUCATION

Bachelor of Science in Electrical and Computer Engineering | GPA: 3.85 / 4.0

State University | Graduated May 2024

## PROFESSIONAL EXPERIENCE

Silicon RTL Design Intern | Apex Microelectronics Inc. (June 2023 - Dec 2023)

- Architected synthesizable Verilog modules for a multi-master SPI bus controller.
- Optimized Finite State Machine (FSM) transitions, reducing critical path latency by 32%.
- Resolved multi-bit Clock Domain Crossing (CDC) metastability violations using dual-clock Gray code FIFOs.
- Wrote automated regression testbenches in Python/ModelSim, increasing code coverage to 96.5%.

## ACADEMIC & TECHNICAL PROJECTS

Hardware-Accelerated Neural Network Core on Xilinx Artix-7 FPGA

- Designed pipelined 8-bit MAC units in Verilog capable of 120 MHz peak operating frequency.
- Achieved complete timing closure and reduced BRAM utilization by 22% using custom memory banking.

Full-Duplex UART Communication Engine with Parity Generator

- Implemented baud rate generator with configurable oversampling (16x) and framing error detection.

## ACHIEVEMENTS & CERTIFICATIONS

- First Place Winner, Annual IEEE Digital VLSI Circuit Design Hackathon
- Certified in Advanced SystemVerilog Verification and Static Timing Analysis (STA)